2.5: Value at Risk (VaR) and Expected Shortfall (ES)

In Sections 2.3 and 2.4, we defined volatility as a measure of dispersion in returns and demonstrated that volatility is time-varying and persistent.

However, volatility measures **overall variability**, not specifically extreme downside risk.

In practice, financial institutions care deeply about **tail risk**:

- How large could losses be?
- How likely are extreme losses?
- How much capital must be held to survive adverse scenarios?

Two widely used measures answer these questions:

- Value at Risk (VaR)
- Expected Shortfall (ES)

These measures focus explicitly on the **left tail** of the return distribution.

Watch

Value at Risk

Details Transcript

0:00/0:00

R Script (https://learn.usc.edu.au/courses/35359/files/3294296?wrap=1) (https://learn.usc.edu.au/courses/35359/files/3294296/download?download_frd=1)

Value at Risk (VaR)

Value at Risk answers the following question:

What loss level will only be exceeded with a specified small probability over a given time horizon?

For example,

A 95% one-day VaR of −3.5% means there is a 5% probability that the asset will lose more than 3.5% in one day.

If the portfolio is worth $10 million, this means:

There is a 5% probability that the portfolio will lose more than $350,000 in one day.

Important: Sign Convention

Returns can be positive or negative.

- Losses correspond to negative returns.
- VaR in return terms will therefore usually be a **negative number**.

If the 5th percentile of returns is -3%, this means:

- 5% of returns are worse than -3%.
- There is a 5% probability of losing more than 3% in one day.
- There is a 95% probability that the return will be greater than -3%.

If converting to dollar terms, we multiply by portfolio value and take the magnitude.

Computing Historical VaR in R

We now compute VaR using **historical simulation**, the most transparent approach.

Step 1: Download Apple daily prices

```
library(tidyverse)

library(tidyquant)

aapl_daily <- tq_get("AAPL", from = "2000-01-01")
```

This dataset contains a column called adjusted, which accounts for dividends and splits.

Step 2: Convert prices into daily returns

```
aapl_daily_ret <- aapl_daily %>%

  tq_transmute(

    select = adjusted,

    mutate_fun = periodReturn,

    period = "daily",

    type = "arithmetic" # I invite you to experiment with "log" instead of "arithmetic"

  )
```

The output contains:

- date
- daily.returns

Before you proceed, check the output.

Step 3: Compute 95% VaR

```
aapl_var_95 <- quantile(aapl_daily_ret$daily.returns, 0.05)

aapl_var_95
```

Explanation:

- quantile() computes percentiles of a numeric vector.
- aapl_daily_ret$daily.returns extracts the return column.
- 0.05 means the 5th percentile.
- The result is stored in aapl_var_95.

If the result is:

-0.0356

Interpretation:

- The 5th percentile daily return is -3.56%.
- There is a 5% probability that the daily return will be worse than -3.56%.
- Equivalently, there is a 5% probability of losing more than 3.56% in one day.

Converting to Dollar VaR

Suppose portfolio value is $200 million.

Dollar VaR = 0.0356 × 200,000,000
Dollar VaR = -$7120000

Interpretation:

There is a 5% probability of losing more than $7.12 million in one day.

Limitations of VaR

VaR has an important weakness:

It tells us the cutoff point, but not how severe losses are beyond that cutoff.

For example:

- VaR = -3.56%
- But losses could be -4%, -8%, or -15%
- VaR does not distinguish between these cases

Expected Shortfall

Expected Shortfall answers: If we are in the worst 5% of outcomes, what is the average loss?

Formally: Expected Shortfall is the mean return conditional on being below the VaR threshold.

Compute Expected Shortfall in R

```
aapl_es_95 <- aapl_daily_ret %>%

  filter(daily.returns <= aapl_var_95) %>%

  summarise(es = mean(daily.returns))

aapl_es_95
```

Line-by-line explanation:

- filter(daily.returns <= aapl_var_95) keeps only the worst 5% of returns.
- summarise() computes the mean of those extreme returns.

- The result is stored as aapl_es_95.

If the result is:

-0.054

Interpretation:

Among the worst 5% of days, the average loss was 5.4%.

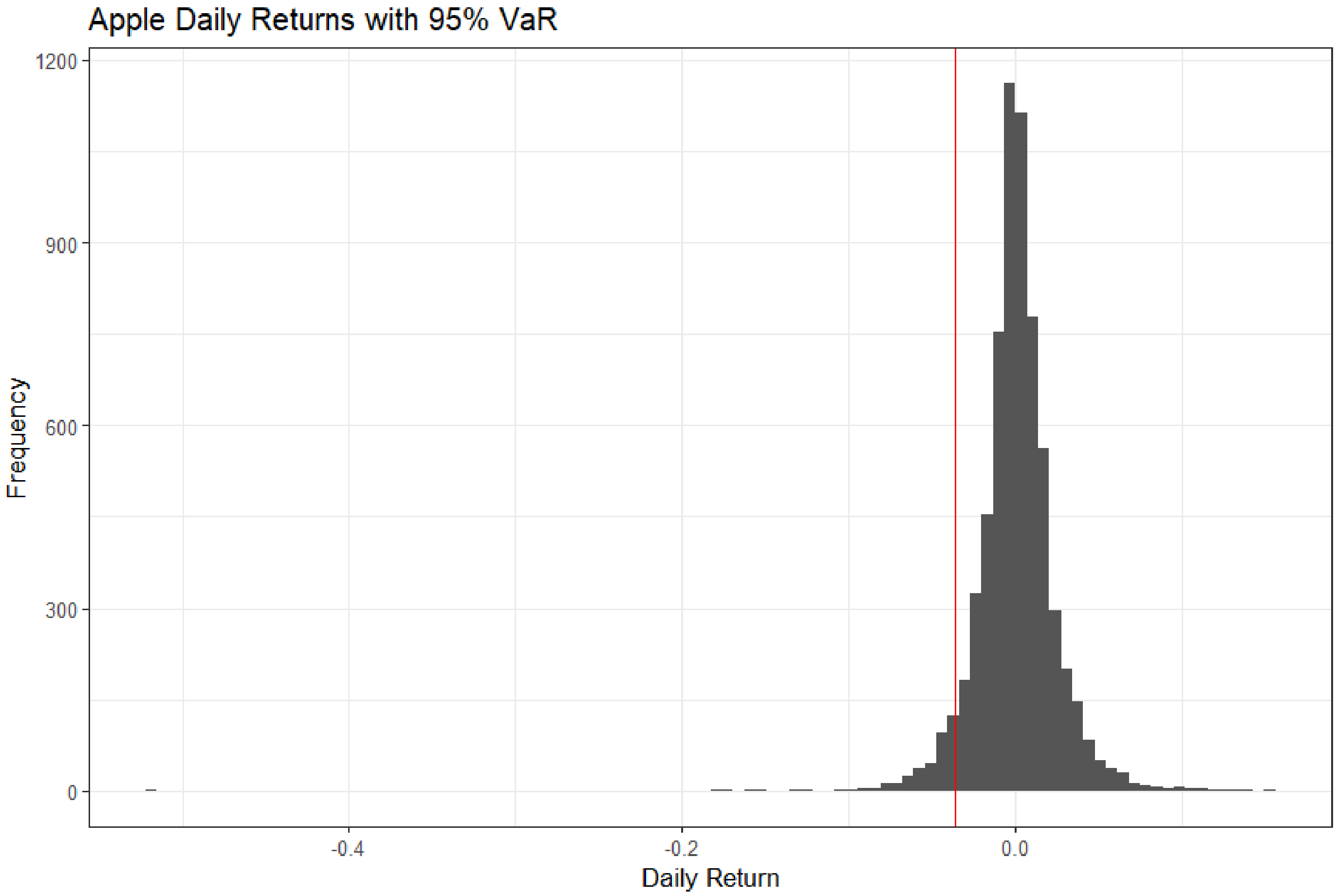
Expected Shortfall is always:

- More negative than VaR (in return space)
- More informative about tail severity

Visualising VaR

We can visualise the cutoff.

```
ggplot(aapl_daily_ret, aes(x = daily.returns)) +

 geom_histogram(bins = 100) +

 geom_vline(xintercept = aapl_var_95, colour = "red") +

 labs(

  title = "Apple Daily Returns with 95% VaR",

  x = "Daily Return",

  y = "Frequency"

 ) +

 theme_bw()
```



Explanation:

- geom_histogram() shows the distribution of returns.
- geom_vline() draws a vertical line at VaR.
- The red line marks the 5% cutoff.

Everything to the left of that line represents extreme losses.